Thai Dang Khoa Tran

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PROFESSIONAL SUMMARY

Software engineer and MS Computer Science student (graduating December 2026) with 2+ years of professional full-stack experience building production systems for major Japanese retail chains. Now work mainly in TypeScript and Python, building data pipelines, serverless AWS applications, and RAG/LLM features, with automated tests at every level.

EDUCATION

University of the Pacific, Stockton, CA January 2025 - Present
MS Computer Science | GPA: 3.86 | Expected graduation: December 2026

University of Science, VNU - HCM, Vietnam 2018 - 2022
BS Information Systems

EXPERIENCE

Full-Stack Software Engineer November 2022 - December 2024
UNICCS Co., Ltd — retail software provider; #1 core-system share among Japan's top 30 drugstore chains

- Architected full-stack features for TAURUS, a core retail ERP (purchase ordering, SKU-level inventory, departmental P&L) used by Don Quijote, Welcia, and Sundrug, with Java Spring Boot backends and JavaScript/Vue.js frontends.
- Reduced query response times by 40% through SQL tuning, indexing, and schema redesign on PostgreSQL and DB2.
- Cut a dashboard stored procedure's runtime from 10 minutes to 30 seconds by isolating the slow join and indexing the key field.
- Deployed and maintained services on AWS, automating build, deployment, and monitoring workflows with Shell scripts.

Tech: Java Spring Boot, JavaScript, Vue.js, PostgreSQL, DB2, SQL, AWS, Shell, HTML, CSS

Software Engineering Intern December 2021 - June 2022
FPT Telecom — part of FPT Corporation, one of Vietnam's largest tech companies

- Built Node.js/Express REST APIs for real-time monitoring of POP stations across FPT Telecom's Southern Vietnam network.
- Developed React/Next.js dashboards with Highcharts, backed by MongoDB schemas designed for fast monitoring queries.

Tech: React.js, Next.js, Node.js, Express, MongoDB, Highcharts

SKILLS

- **Languages:** JavaScript, TypeScript, Python, Java, C++, C#, SQL, HTML, CSS
- **Frameworks:** React.js, Next.js, Vue.js, Node.js, Express, Spring Boot, FastAPI, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, SQL Server, ClickHouse, Redis
- **Testing & CI/CD:** Vitest, Playwright, Unit testing, Integration testing, End-to-end testing, GitHub Actions
- **Cloud & Tools:** AWS (Lambda, EventBridge, S3, CloudFront, SNS, DynamoDB, IAM, SAM), Docker, Git, GitHub, Claude Code
- **AI/ML:** PyTorch, Hugging Face Transformers, BERT fine-tuning, NLP, ONNX Runtime, RAG, LLM APIs (Groq)
- **Concepts:** OOP, Data Structures & Algorithms, RESTful APIs, Database Design, Data Pipelines/ETL, Serverless, Infrastructure as Code, Agile/Scrum, LLM Integration, Prompt Engineering, AI Agents

PROJECTS

US Weather Forecast Pipeline — *Personal Project* Live App · Source Code

- Built an end-to-end data pipeline (NWS + Open-Meteo → MongoDB → ClickHouse → Redis → Hono API → React) processing 1.8M+ hourly observations and 4.5M+ forecast snapshots for 55 US cities since 2023, with incremental loads and data-quality checks.
- Deployed the live side on AWS as infrastructure as code (SAM): Lambda + EventBridge collection and publishing, S3 + CloudFront delivery, SNS alert emails with public sign-ups, and least-privilege IAM, all at $0 on the free tier.
- Comparing 4 weather models' accuracy in an interactive dashboard with map drill-down, forecast replay, and a live status page.
- Used Claude Code to build and compare 3 versions of 5 key features, keeping the best (Redis caching cut p95 latency 12×); fixed SQL/NoSQL-injection and IAM security flaws; 400+ automated tests in CI.

Tech: Node.js, Hono, TypeScript, React, D3, MongoDB, ClickHouse, Redis, Docker, AWS (Lambda, S3, CloudFront, DynamoDB, SAM), Vitest

RAG-Assisted Natural Language to SQL System — *Personal Project* Live App · Source Code

- Built and deployed a full-stack RAG assistant (React + FastAPI) that turns plain-English questions about any CSV into SQL, using a fine-tuned BERT intent classifier, term-gated retrieval, and gpt-oss LLMs with fallbacks.
- Raised accuracy on questions that need business definitions from 0% to 95% on a held-out domain and diagnosed why naive few-shot RAG lowered accuracy.
- Built a SELECT-only SQL safety layer that blocked 13 prompt-injection variants; all 120B benchmark runs cost under $0.25.

Tech: Python, FastAPI, React, TypeScript, BERT, ONNX Runtime, Groq, SQLite, pytest, Vercel, Render

CineVibes — *Personal Project* Live App · Source Code

- Built a full-stack movie community where fans chat in spoiler-aware, category-based threads and follow trending films.
- Added CineBot, an AI assistant (Groq LLM) that recommends movies and answers trivia from the app's own catalog.

Tech: React, Node.js, Express, MongoDB, Vite, JWT Auth, Groq, Vitest, Playwright, GitHub Actions